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Software Engineering

Requirements

COMP1003 - Lecture 3

Dr Horia A. Maier



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Meme of the Week



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Today's lecture

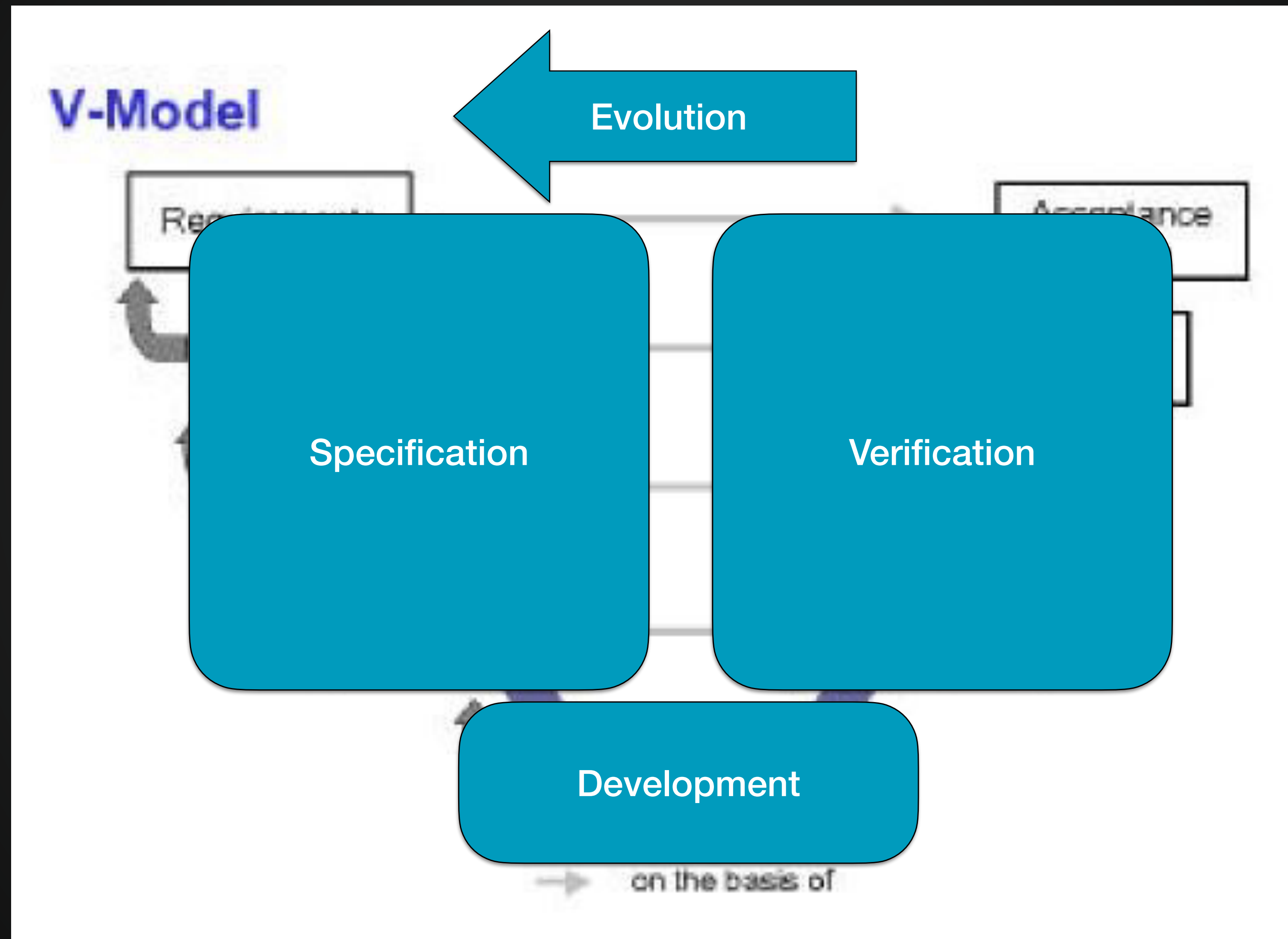


Core SE Processes

- **Specification** - designing the system for what the customer wants
- **Development** - production of the software system
- **Validation** - checking that the software is what the customer wanted
- **Evolution** - changing code in response to new requirements

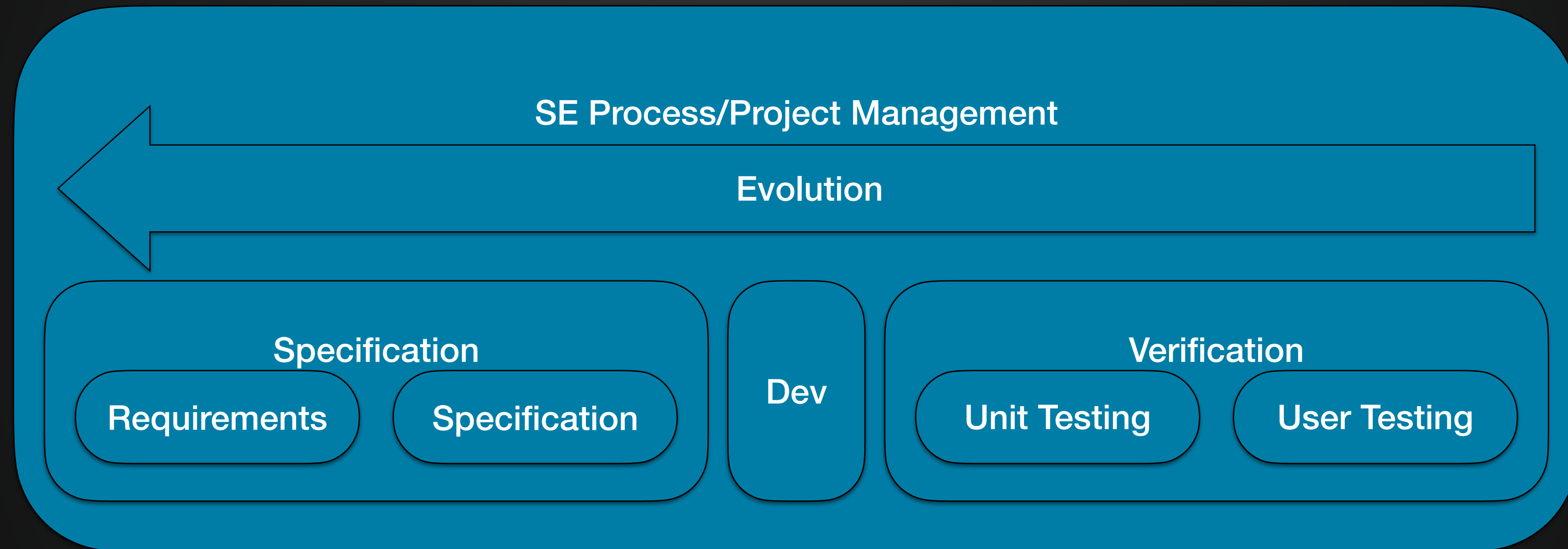


Keeping Track of This Module



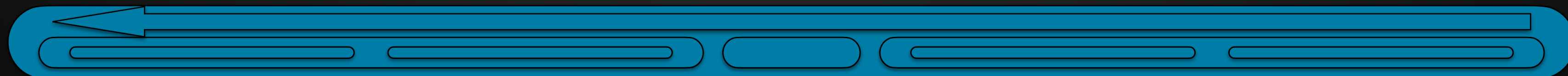


Keeping Track of This Module



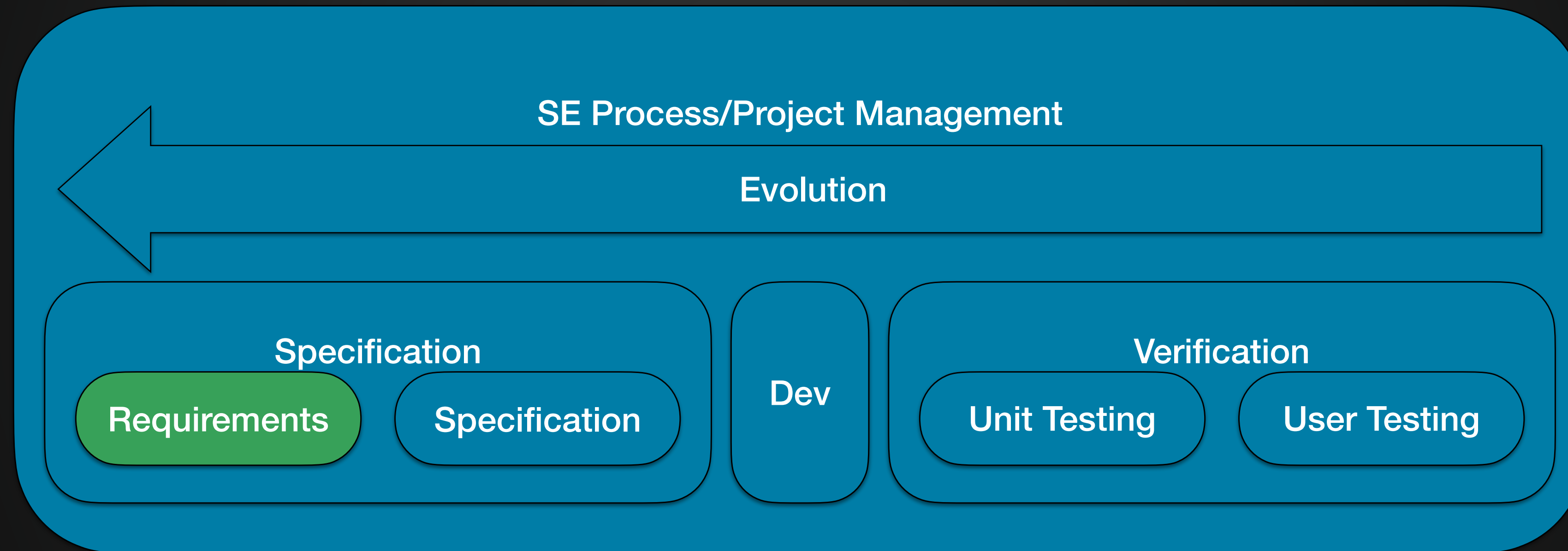


Keeping Track of This Module



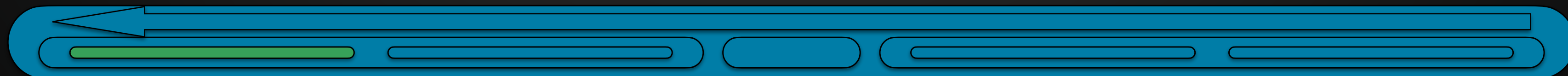


Keeping Track of This Module





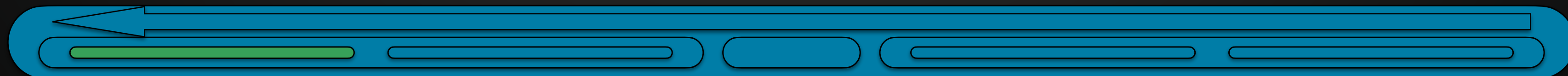
Keeping Track of This Module





Today's Learning Outcomes

- 1) Why Requirements are so important
 - That Over-Budget costs are **mostly** associated with Bad Requirements
- 2) What Requirements Engineering is
- 3) Introduction to Requirements Elicitation stage
- 4) Initial Methods for identifying requirements areas





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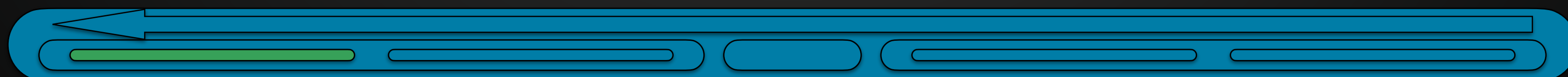
Why requirements are important



Ongoing SE Project Failures

- The global cost of IT failure was estimated at + 6 trillion dollars in 2009
Source: Roger Sessions, The IT Complexity Crisis: Danger and Opportunity
- Only 32% of software projects were “successfully completed” (i.e., on time, on cost, and with expected functionality) in 2009
Source: Standish CHAOS 2009 Update
- Only 16% of software projects were successfully completed in the UK in 2009
Source: British Computer Society
- “A failing industry ...
 - *If building engineers built buildings with the same care as software engineers build systems, the first woodpecker to come along would be the end of civilization as we know it.”*

Source: Paul Dorsey, Top 10 Reasons Why Systems Projects Fail





Reported in 2023

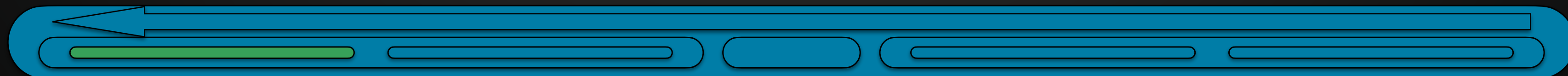
- 31.1% of software projects will be canceled before they are completed.
- Only 16.2% of software projects are completed on time and on budget.
- Poor requirements are the leading cause of project failure at 39.03%.
- 57% of the projects that fail are due to communication breakdown.
- 29% of software projects fail due to insufficient or poor testing.

<https://zipdo.co/statistics/software-project-failure/>



What causes failure?

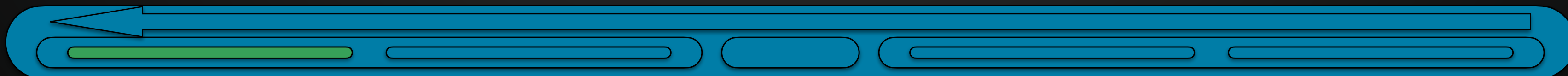
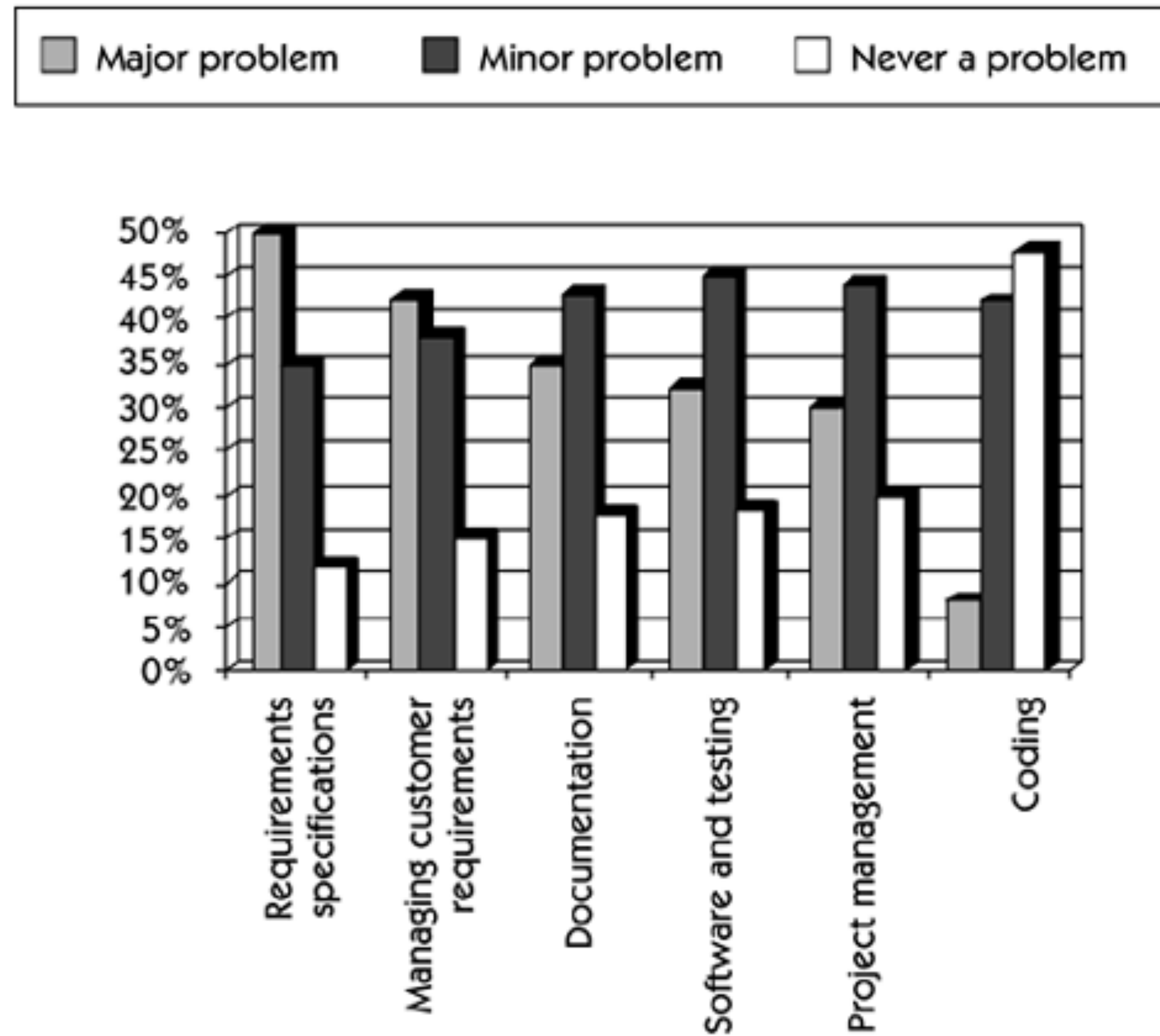
- **Incomplete requirements** and **lack of user involvement** rank top of the list of causes of project failure
 - Source: Standish CHAOS 1995 report
- Understanding product requirements is **the major problem** in software development
- Coding or programming is not a major problem





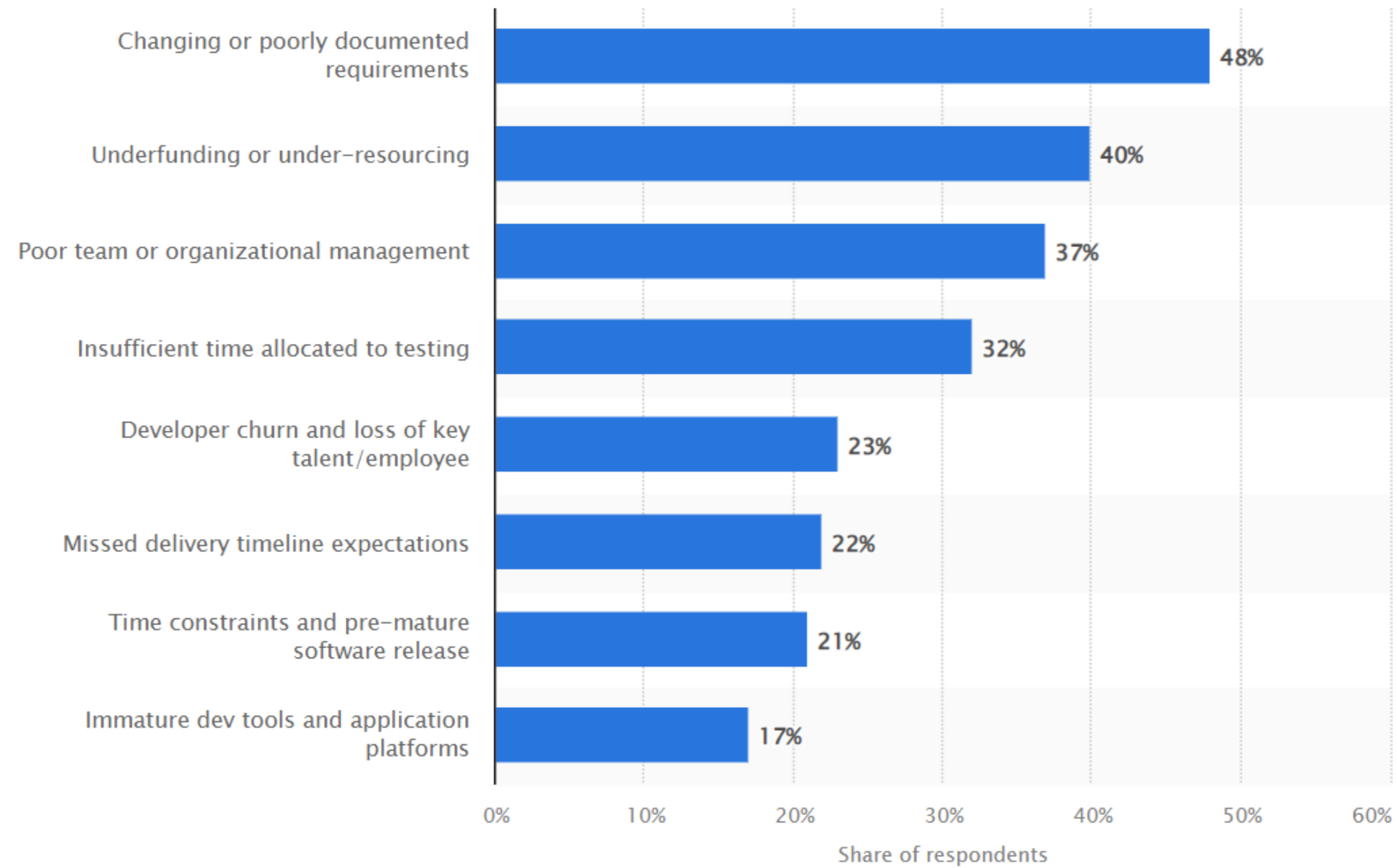
What causes failure?

Figure 1-1. Largest software development problems by category. (Data derived from [ESPITI \[1995\].](#))





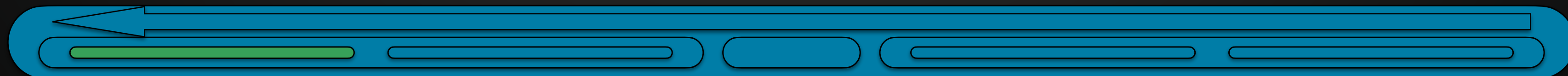
From 2015 report





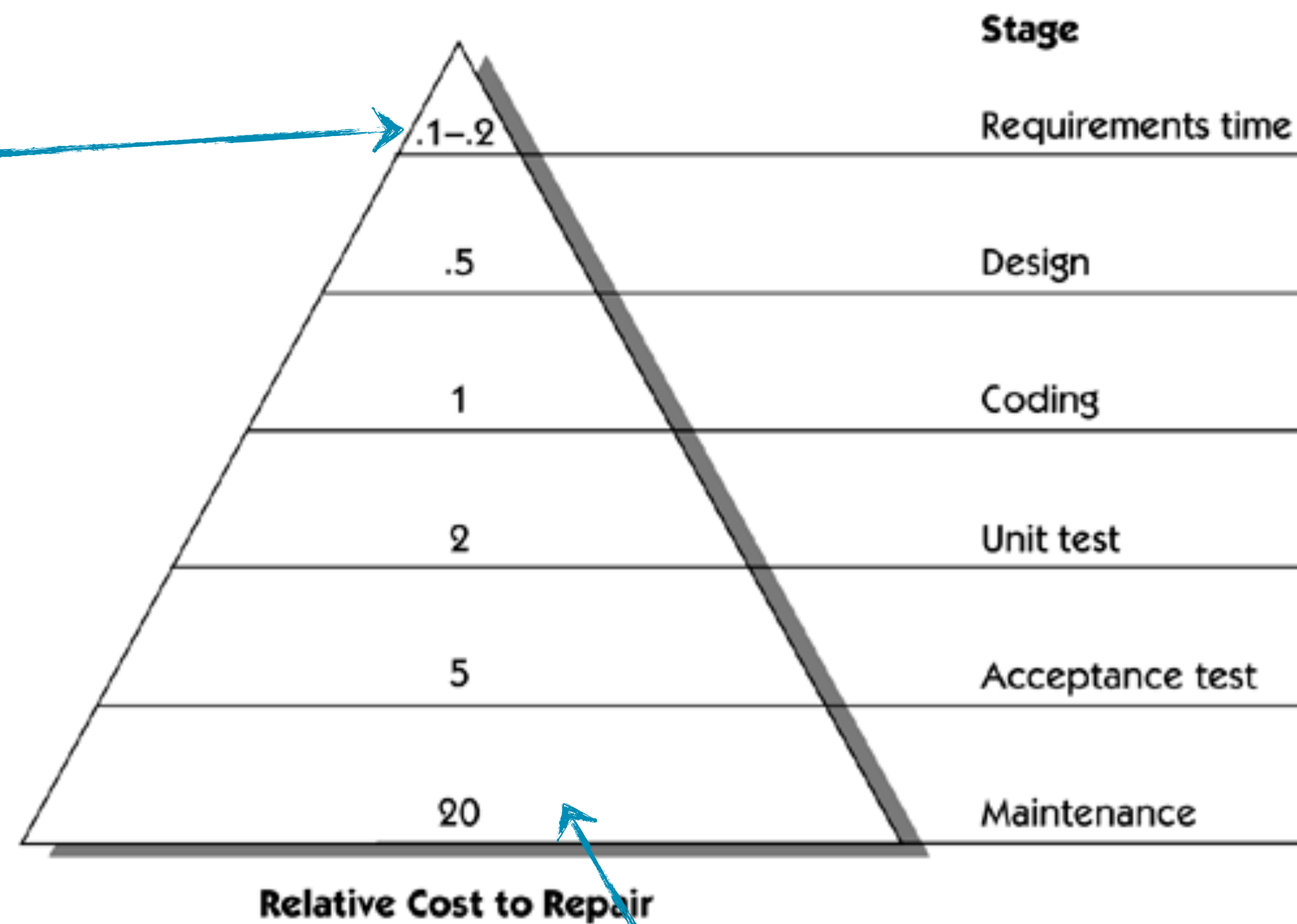
What causes failure?

- “**The hardest single part of building a software system is deciding precisely what to build** ... No other part of the work so cripples the resulting system if done wrong. No other part is more difficult to rectify later.”
 - Source: Fredrick Brooks, No Silver Bullet: Essence and Accidents of Software Engineering
- As much as a **200:1 cost savings** results from finding errors in the requirements stage versus finding errors in the maintenance stage of the software lifecycle



Cost of Fixing Defects

Figure 1-2. Relative cost to repair a defect at different lifecycle phases. (Data derived from [Davis \[1993\]](#).)



We want to fix problems here (obviously)

Basically, you want to avoid changes after release



Cost of Fixing Defects

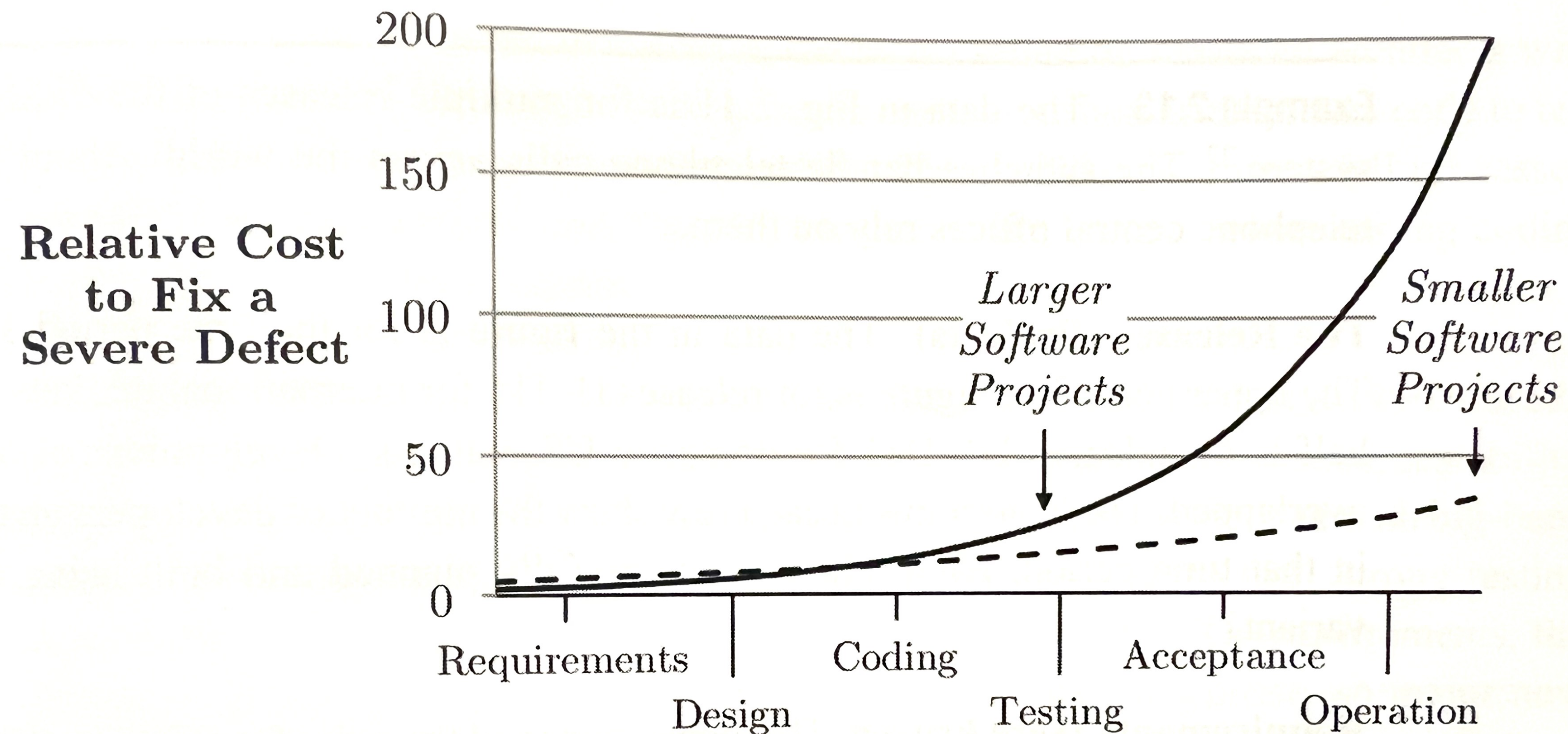


Figure 2.10 For severe defects, the later the fix, the greater the cost. (The original diagram had a log scale for the relative cost.)

From Textbook



The critical thing

- Is to spend the timing figuring it out properly - **right at the start**



Activity 1

Draw me a design for a bridge





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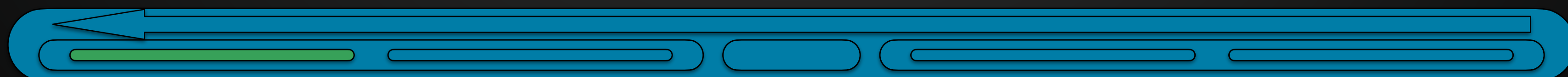
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The problem

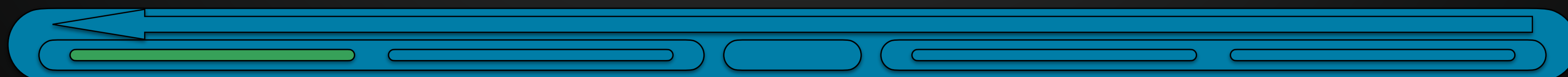


The Requirements problem

- You all want to build software
 - But there is a **great deal more** to doing that successfully than learning to cut code
 - Understanding “stakeholder” requirements (i.e., client, customer, user, and other interested parties needs) is key to effective systems development
 - That means you are going have to learn **how to identify their requirements**
 - Which means you are going to have to learn to work with human beings as well as machines ...



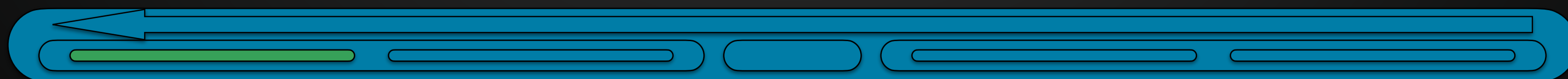
The Requirements problem





The Requirements problem

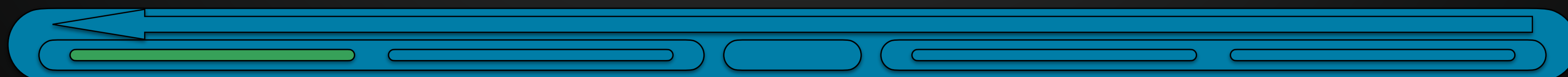
- *“I would go a step further and assert that it is really impossible for clients, even those working with software engineers, to specify completely, precisely, and correctly, the exact requirements of a modern software product”*
 - Source: Fredrick Brooks, No Silver Bullet: Essence and Accidents of Software Engineering
- We need to be able to **determine them**, by **working with** clients
- Over the next few lectures we will be learning how to do requirements engineering





The Requirements problem

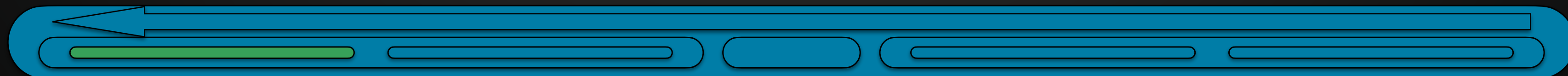
- In more technical terms its about “requirements engineering” or figuring out what **stakeholder needs** and what **services** a product will need to provide **to meet those needs**
- “**Requirements are your project’s foundation.** They define the level of quality you need, facilitate decision making, provide a basis for tracking progress, and serve as the basis for testing.”
- “If you let them remain unstated, you have no opportunity to examine and negotiate them with your customer and no way to tell when your project has met its objectives.”
 - Source: Brian Lawrence, Top Risks of Requirements Engineering





Where do requirements come from?

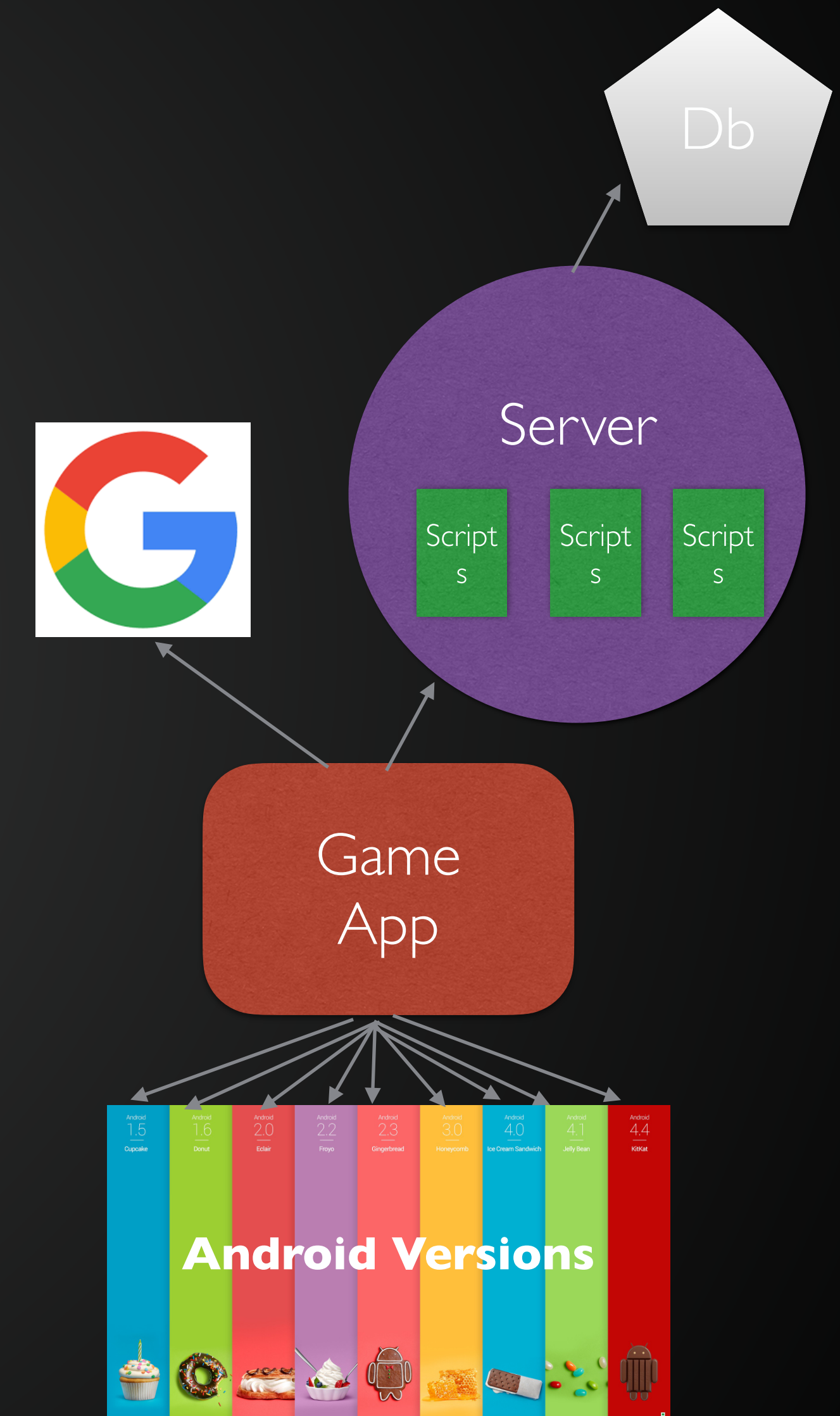
Lets look at our examples



Where do requirements come from?

A phone game app

- What the game idea prescribes
- What we learn from other game apps
- Analysing market segments
- What the platform (app store) requires



Where do requirements come from?

UK Universal Credit (Benefits) Software

- The government designs an invitation to tender
 - Analysis of desired idea
 - Requirements analysis of existing systems
- Companies bid to meet those requirements





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What is Requirements Engineering?

Requirements Engineering

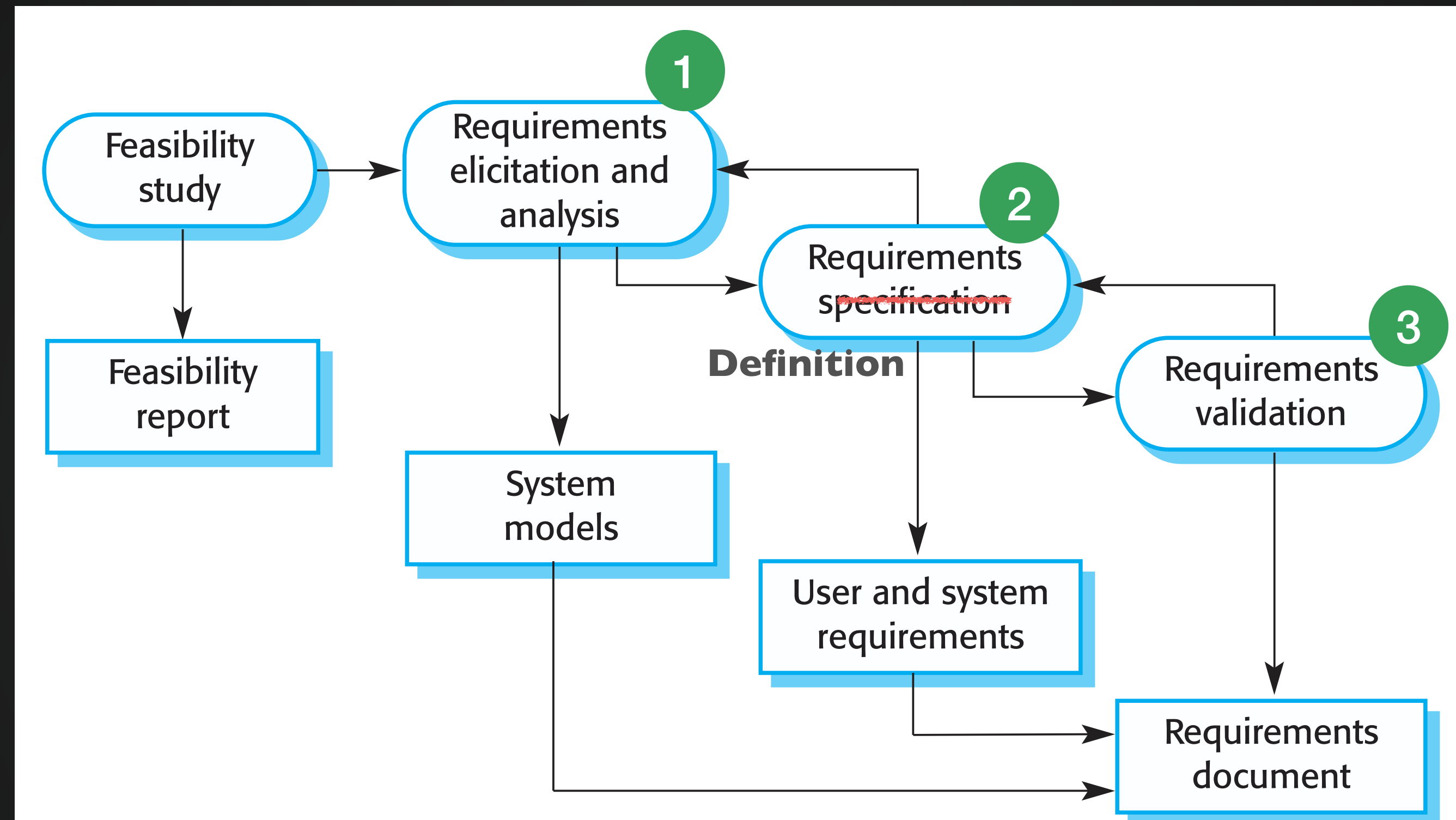


Figure 2.4 The requirements engineering process



Requirements Engineering

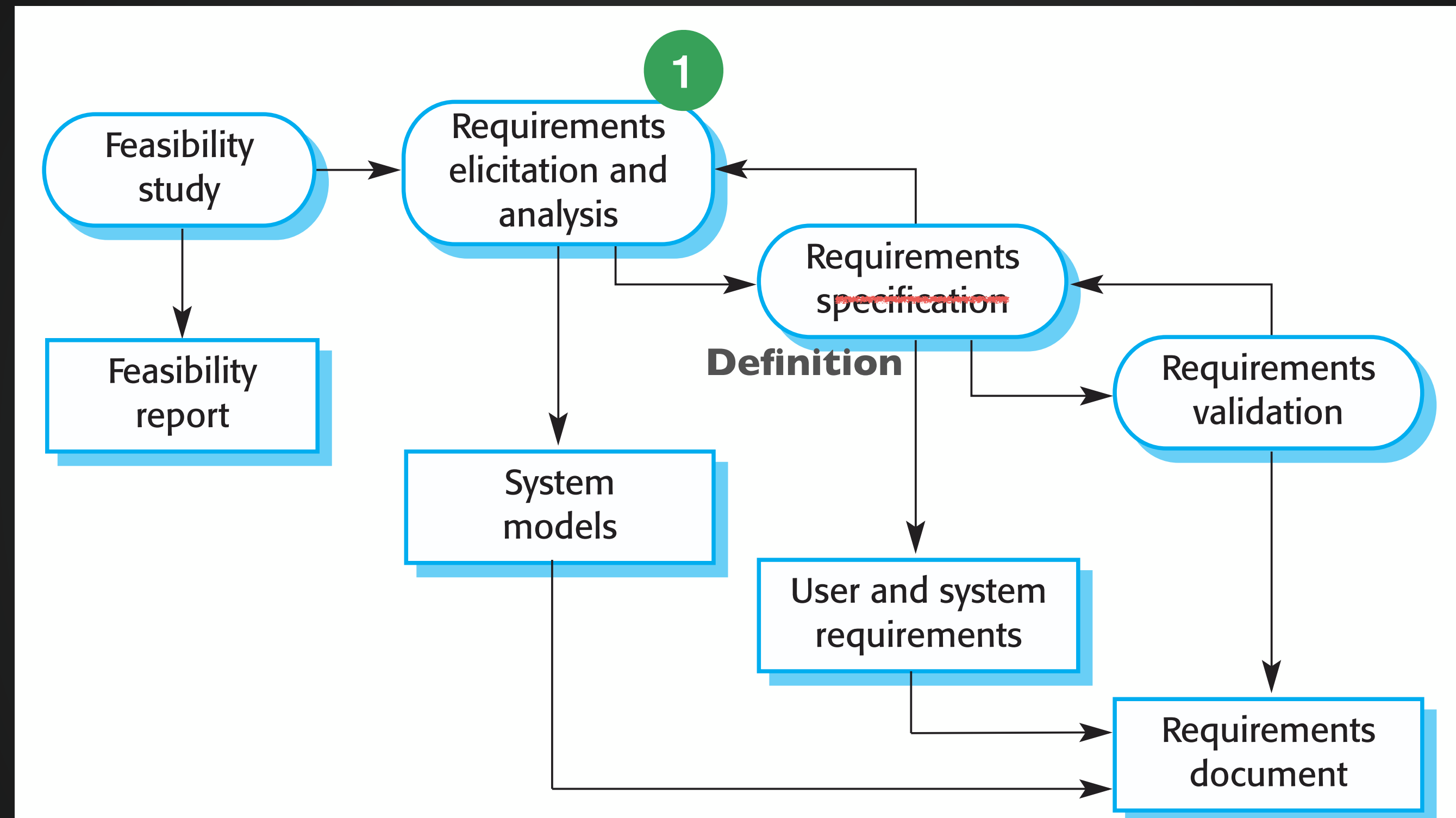


Figure 2.4 The requirements engineering process



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Requirements Elicitation



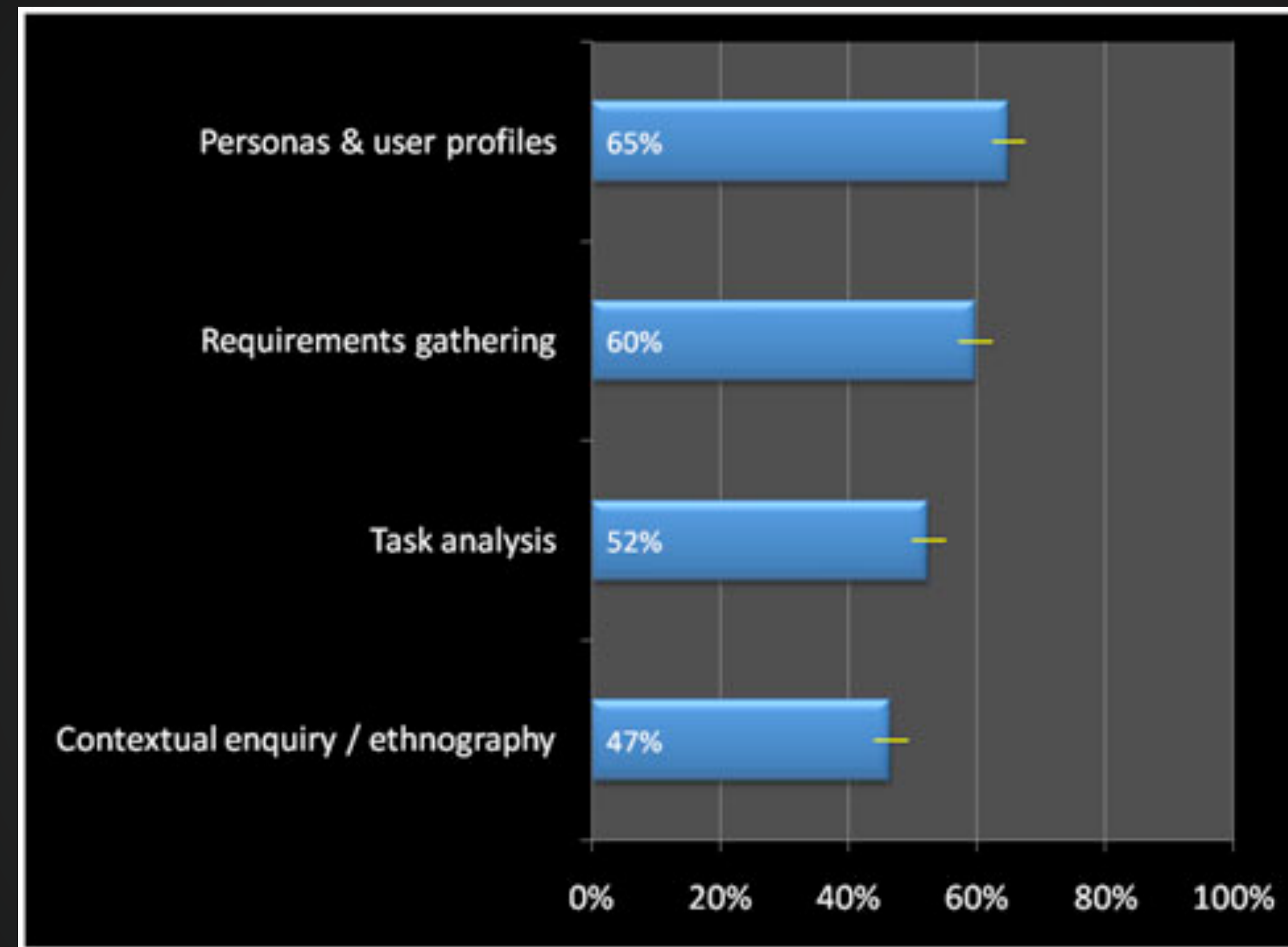
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Initial Methods for Req. Elicitation



Requirements Elicitation



<http://www.measuringusability.com/blog/ux-methods.php>
(survey of thousands of usability professionals)



Stakeholder Analysis

- First stage: determine **all the people** that will use the system
- From a mixture of
 - working with initial briefs
 - problem analysis
 - interviews / discussions
- Have to identify
 - primary stakeholders
 - secondary stakeholders
 - tertiary stakeholders



Stakeholder Analysis - advanced

- There are all sorts of ways of classifying/analysing them (beyond primary, secondary, tertiary)
- Importance/Priority
- Impact of their needs
- You can list example people
- You can list what you want from them



Activity 2

Identify Stakeholders in initial brief

Categorise them as Primary, Secondary, Tertiary



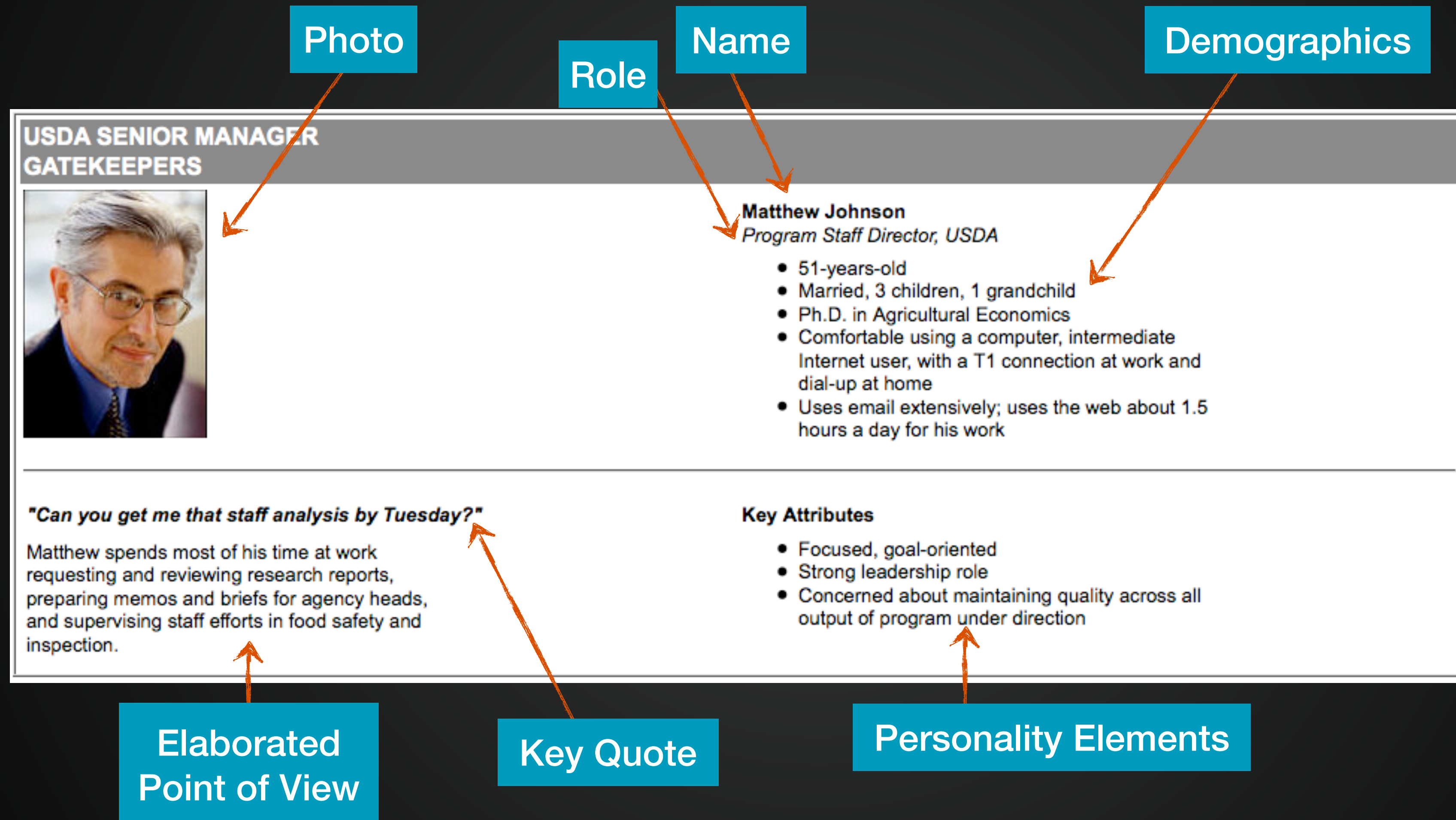
Personas

- Represents a real **type of user** from your stakeholders
- **Main Aim: Personas should differentiate stakeholders clearly**
- Identify: motivations, expectations, goals, knowledge
- Usage: Help you to put yourself in the shoes of...
- Ideally: Try to have a small number - all one page at most



Personas

<http://www.usability.gov/analyze/personas.html>



1

2



Personas - advanced

- Not a report on real people from the client company
- Choose a generic representative name/photo/etc
- Make 2 or 3 that demonstrate **key user types**
- It might be that several stakeholders can be represented by one persona
 - e.g. lecturers / module convenors / supervisors
- It might be that one stakeholder type - needs several personas
 - e.g. regular gamers, casual gamers, high-spenders



Personas - advanced

- “Externalisation”
 - SE mistakes are often from ‘what was not said and not documented’
- At design time: Would ‘Jim’ use this?
 - Stops designers designing for themselves
- Prioritisation of requirements
 - and asking: Does this design satisfy all 3 personas?
- Developing scenarios (next lecture)!



Persona Board

Kivio Users				
	The researcher	The Sysadmin	The OSS developer	The CS student
				
Name	Alexander Weiß	Donald M. Berry	Kristian Larsson	Eric Neville
Age	30	30	26	24
Location	Germany	US	Sweden	France
Social Life	Alexander lives with his girl-friend in a flat in Hamburg.	Donald lives with his wife and 1-year old daughter in a house in Portland.	Kristian shares an apartment with two friends in Stockholm. His girl-friend lives in Uppsala. They see each other every weekend.	Eric lives with his parents in a small city close to Lyon. He visits the university there. Often, he stays at his friend's apartment for playing PC games and programming.
Work Life	He works at centre for environmental systems research and designs plans for replacable energies in a EU-funded project.	He is a lead system administrator in a huge network solutions company in Portland.	A software developer with a dayjob in a medium-sized software company. Works on KDE in his spare time.	He is a student of computer science. Besides university, he performs small programming jobs for people in his neighbourhood.
Computer Experience	All are highly experienced with computers.			
Time at a computer per week	26-50 hours per week	35-50++ hours per week	30-50++ hours per week	25-45 hours per week
Computer tasks	Office tasks and Field-dependent. Also educational and recreational. No development.	Development and network administration. Does not use PC for office tasks, educational, and even recreational.	Mostly development and recreational. Also network administration and office.	Mostly development. Also educational, recreational, and network administration. Does not use for office work.
Relation to OSS.	He is not passionate about OSS.	He is a convinced user of OSS.	He is involved with OSS development.	He is a convinced user of OSS.
Requirements wrt diagramming	office requirements	highest claims	easy-going	eager beaver
Frequency of drawing diagrams	Each 2 nd month	Twice per month	Once a month	Each 2 nd month
Diagram main type	Flowcharts. Also visualising thoughts	All, except sitemap	Visualising thoughts	Diagrams mostly UML

<http://wiki.fluidproject.org/display/fluid/Persona+Format>

Persona Board

TEST CASE APPLICATION PERSONAS				
PERSONA	THINKING	HEARING	SAYING	DOING
DEVELOPER (R)	HAS MY STUFF BROKEN? DID I BREAK THE BUILD IS MY STORY "DONE"		WHAT IS REQUIRED FOR TESTING?	CREATING TEST SUITES, TEST CASES & TESTS UPDATING THEIR OWN TESTS
QA ENGINEER (R)	AM I ON TRACK TO MEET TESTING REQS			EDITING OTHER PEOPLE'S TESTS MAINTAINING RISKS
TEC ADMINISTRATOR				CREATE TESTED ITEMS, TEST TGTS MANAGE CATEGORIES.
QA MGR	"STUFF NEEDS TESTING" "DOES TEST PLAN HAVE ALL TEST CASES" "RISK PROFILE FOR PARTS OF PRODUCT" "IMPLEMENTING RIGHT TYPE OF TESTING"		"HOW MANY TESTS PASSED/FAILED ON A DAILY BASIS" "WHY DID THIS FAIL"	PREPARING TEST PLANS
QA DIRECTOR			WHAT IS THE DELTA CHANGE FROM YESTERDAY	REVIEWING DASHBOARD APPROVING A RELEASE TO GO

<http://scrumprout.tumblr.com/post/30516447738/building-the-persona-board>

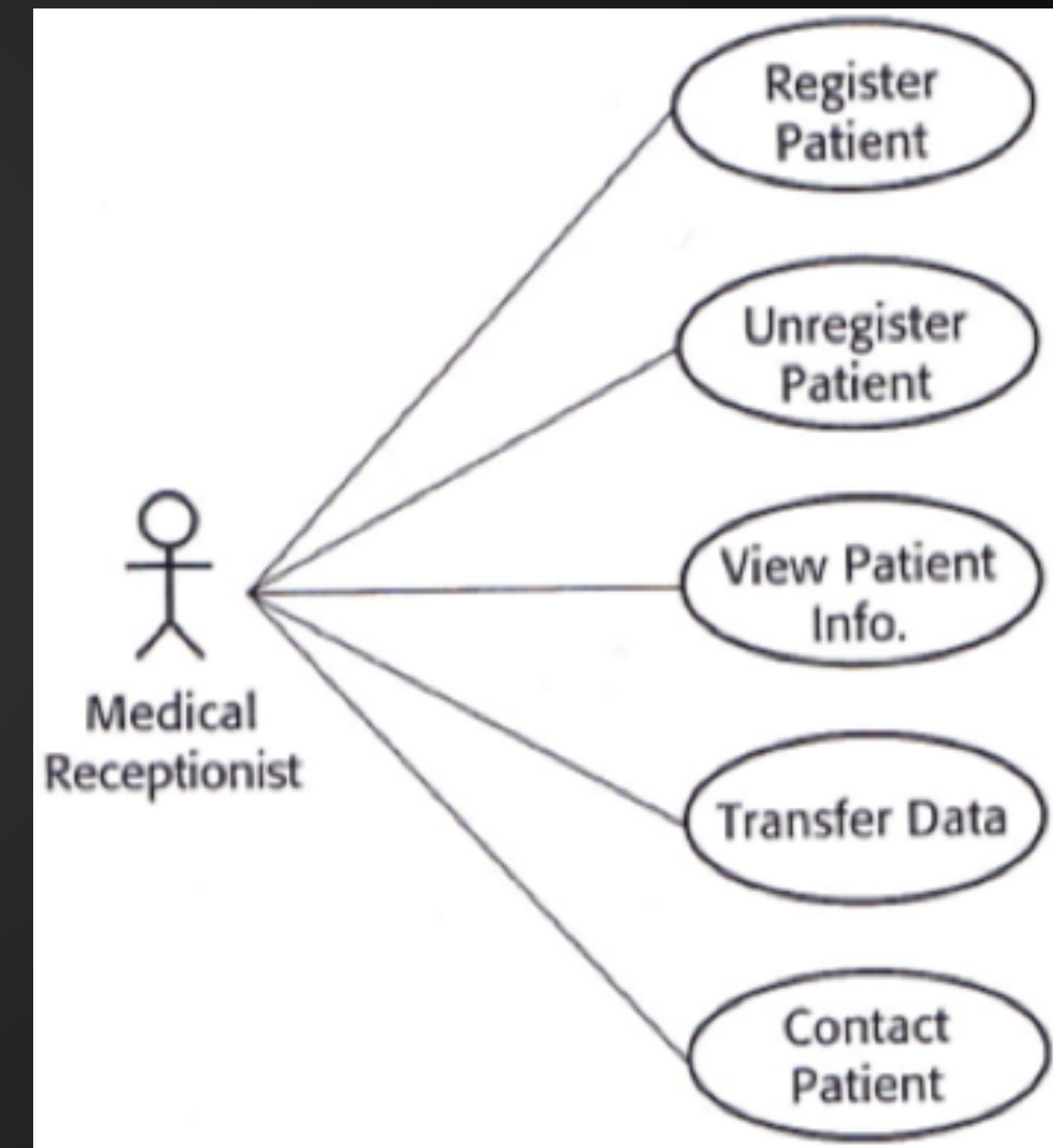


Activity 3

What Personas for our Example?

Use Case Diagrams

- We need to elaborate the tasks that each Stakeholder will do
- The most common method for this is a 'use case' diagram
- They represent the people who use the system and the tasks they have to perform
- We call the people 'actors'



Use Case Diagrams

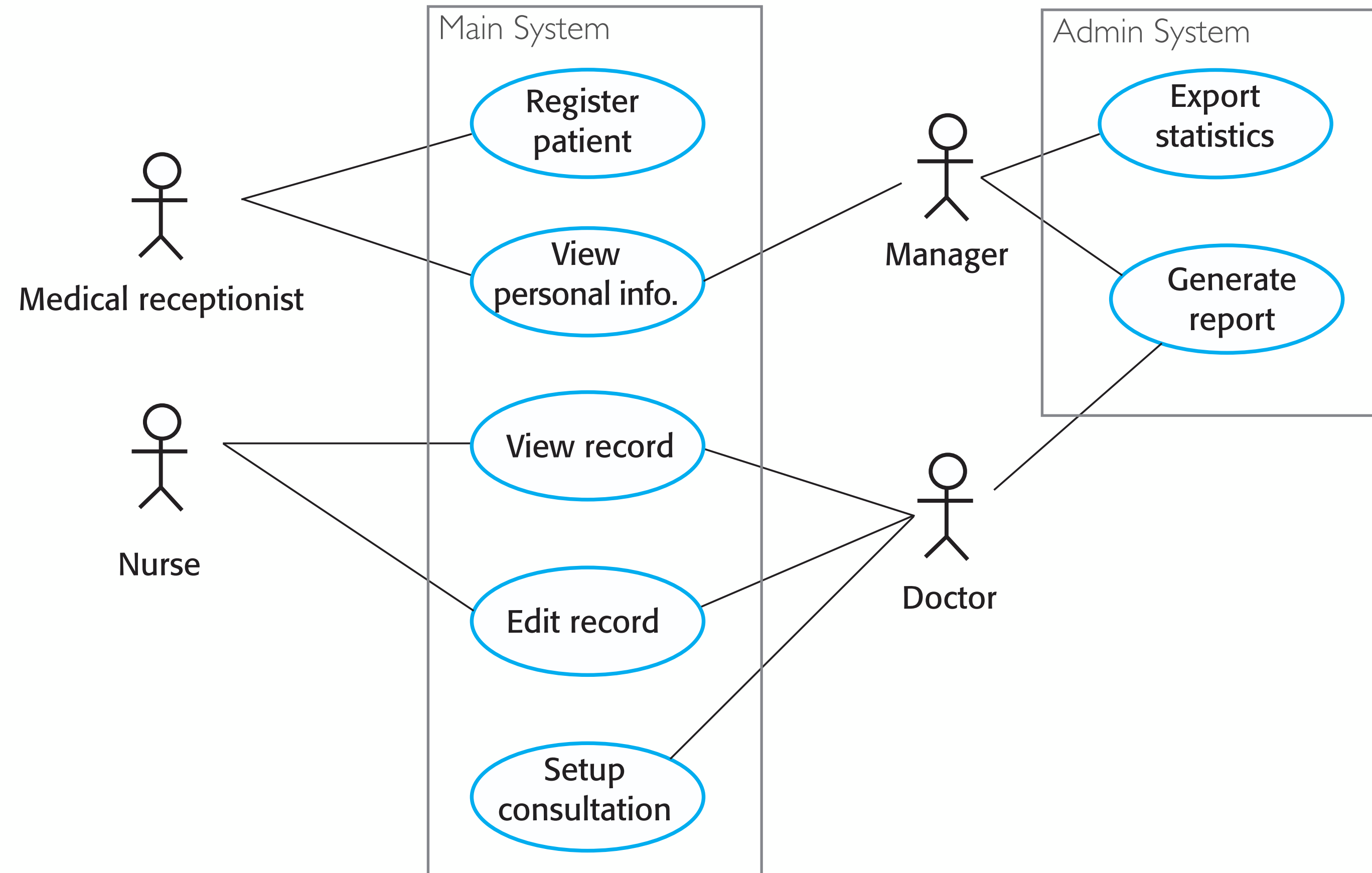
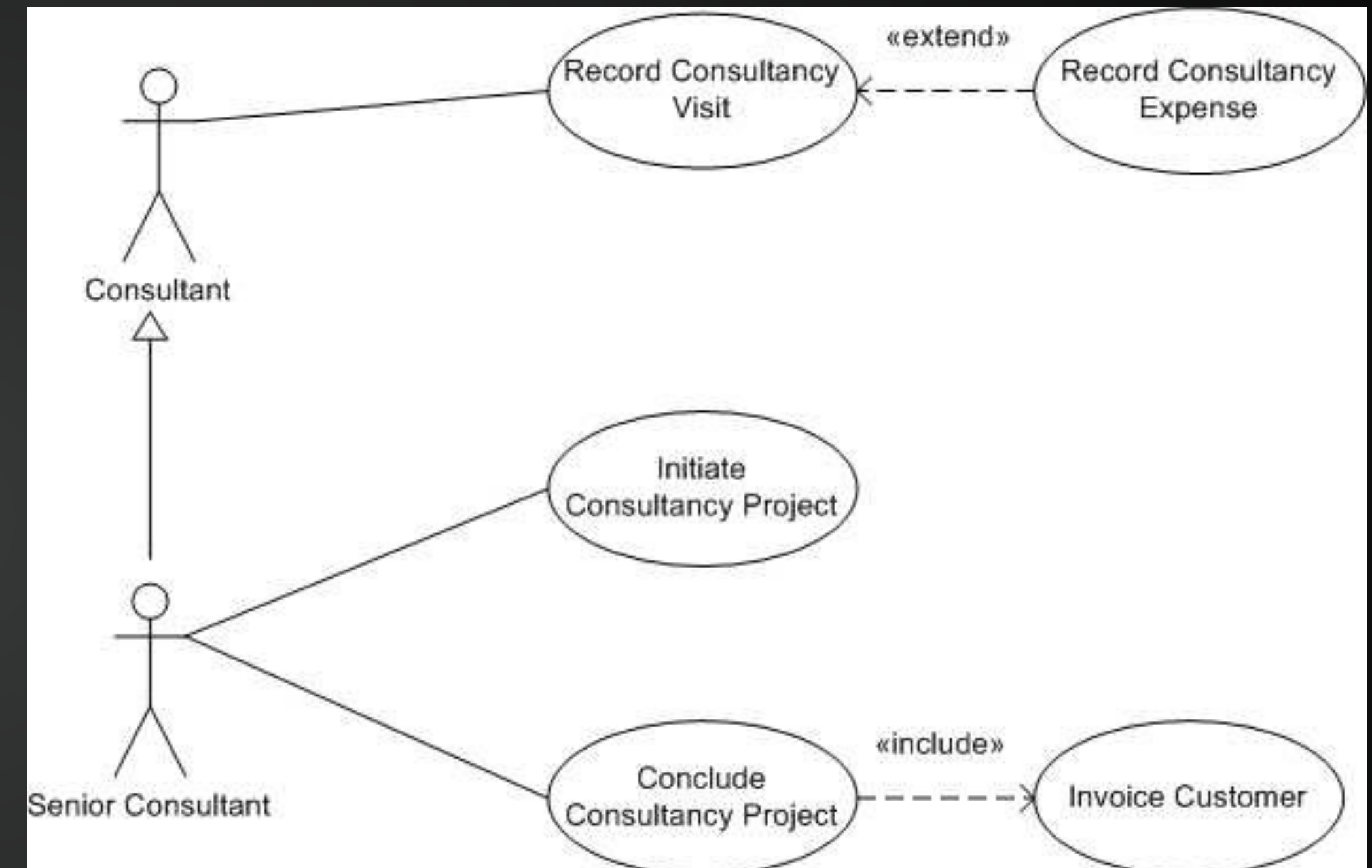


Figure 4.15 Use cases for the MHC-PMS

Use Case Diagrams

- A use case can be an extension of a task
 - (Optional additional things)
- Some tasks may necessarily include other tasks
 - (Mandatory related task)
 - (Often something reused by multiple use cases but cannot be done directly by the user)

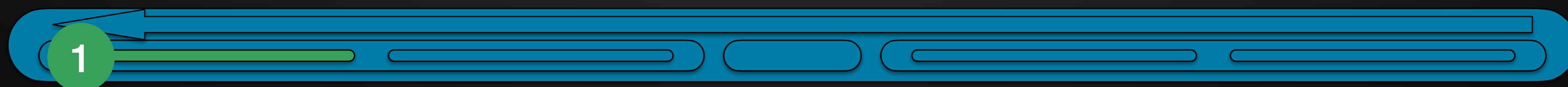


https://geeksworld.com/tutorials/miscellaneous/uml/resources/making_use_case_diagram_using_generalization_in_uml.php



Activity 4

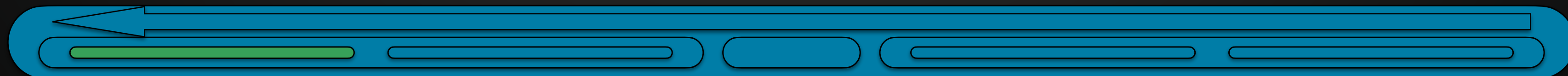
Identify Use Cases for each Stakeholder





Today's Learning Outcomes

- 1) Why Requirements are so important
 - That Over-Budget costs are mostly associated with Bad Requirements
- 2) What Requirements Engineering is
- 3) Introduction to Requirements Elicitation stage
- 4) Initial Methods for identifying requirements areas



DRAFT SCHEDULE			
Week Starting	Lecture & Lab Slots		
	Monday 11-12noon Business B52	Tuesday 5-6pm Business B52	Wednesday 11-1pm CS Atrium, CS A07, CS A32
1) 26 Jan	Module Intro	Git Ready for Teams	No Lab
2) 2 Feb	Requirements Engineering	Requirements Gathering	Team Activity: Git and Markdown
3) 9 Feb	Requirements Models	Requirements Validation	Team Activity: UML Intro
4) 16 Feb	Specification	Prototyping	Team Activity - More UML
5) 23 Feb	Code Planning	Coding in Teams	Team Activity - Requirements Gathering
6) 2 Mar	Test-Driven Development & JUnit Demo	Release Testing	Team Activity - JUnit
7) 9 Mar	Config & Deployment	Advanced Git Demo	Team Activity - Advanced Git
8) 16 Mar	Software Maintenance	Industry Guest Lecture?	CW start assessed 50% of the module (18/03/2026)
9) 23 Mar	Overview 1: Traditional vs Agile	Overview 2: Software Quality	CW support continued
E A S T E R			
10) 27 Apr	Overview 3: Risk Management	Project Planning & Management	CW Support continued
11) 4 May	Bank Holiday (No Lecture)	No Lecture	CW Delivery Wednesday the 6th of May 2026 at 5pm (ideally end of lab)
12) 11 May	TBC	No Lecture	No Lab